Package ‘fracreg’

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Type Package

Title Fractional Regression Models

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Description A comprehensive toolkit for the estimation and specification analysis of fractional regression models. Provides advanced routines for fitting univariate one-part, hurdle two-part, and double-inflated three-part fractional models. Further incorporates estimators for panel data settings and addresses unobserved heterogeneity and endogeneity via correlated random effects and control function approaches. Calculates analytical partial effects across all model types and includes generalized goodness-of-functional-form (GGOFF) and RESET hypothesis tests.

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NeedsCompilation no

Repository CRAN

URL https://SulmanOlieko.github.io/fracreg

Suggests knitr, rmarkdown

VignetteBuilder knitr

Contents

`fracreg-package` *Fractional Regression Models*

Description

Provides comprehensive tools for the estimation and specification analysis of fractional regression models. It supports univariate one-part, hurdle two-part, and double-inflated three-part models. The package also incorporates estimators for panel data settings (CRE, GMM, QML) and addresses unobserved heterogeneity and endogeneity via correlated random effects and control function approaches. It supports various link functions, calculates average and conditional partial effects analytically across all model types, and provides robust specification tests such as RESET, P test, and GGOFF tests.

Details

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Type:	Package
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Author(s)

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References

Papke, L. E. and Wooldridge, J. M. (1996), "Econometric methods for fractional response variables with an application to 401(k) plan participation rates", *Journal of Applied Econometrics*, 11(6), 619-632.

Ramalho, E.A., J.J.S. Ramalho and J.M.R. Murteira (2011), "Alternative estimating and testing empirical strategies for fractional regression models", *Journal of Economic Surveys*, 25(1), 19-68.

Ramalho, E. A., Ramalho, J. J. S., and Murteira, J. M. R. (2014), "A two-part fractional regression model for the spatial distribution of vineyards in Portugal", *Journal of Applied Econometrics*, 29(4), 607-630.

Fang, K., & Ma, S. (2013), "Three-part model for fractional response variables with application to Chinese household health insurance coverage", *Journal of Applied Statistics*, 40(5), 925-940.

`fracreg` *Fitting Fractional Regression Models*

Description

`fracreg` is used to fit fractional regression models, which are appropriate for responses that are proportions, percentages, or fractions restricted to the [0, 1] interval. It supports standard one-part models, two-part hurdle models for modeling boundary values at 0 or 1, and three-part models for double inflation at both 0 and 1.

Usage

```
fracreg(y, x, x2 = x, linkbin, linkfrac, type = "1P", inflation = 0,
        intercept = TRUE, table = TRUE, variance = TRUE, var.type = "default",
        var.eim = TRUE, var.cluster, dfc = FALSE, ...)
```

Arguments

`y`	a numeric vector containing the values of the response variable.
`x`	a numeric matrix, with column names, containing the values of the covariates.
`x2`	a numeric matrix, with column names, containing the values of the covariates in the fractional component of two-part models if option `type = "2P"` is defined. Defaults to `x`.
`linkbin`	a description of the link function to use in the binary component of a two-part fractional regression model, or a vector of two link functions for the two binary components of a three-part model (e.g. `c("logit", "probit")`). Available options: `logit`, `probit`, `cauchit`, `loglog`, `cloglog`.
`linkfrac`	a description of the link function to use in standard fractional regression models or in the fractional component of a two-part fractional regression model. Available options: `logit`, `probit`, `cauchit`, `loglog`, `cloglog`.
`type`	a description of the model to estimate: a standard one-part model (`1P`, the default), a two-part model (`2P`), the binary component of a two-part model (`2Pbin`), the fractional component of a two-part model (`2Pfrac`), or a three-part model (`3P`) for double boundary inflation.
`inflation`	a numeric value indicating which of the extreme values of `0` (the default) or `1` is the relevant boundary value for defining two-part fractional regression models.
`intercept`	a logical value indicating whether the model should include a constant term or not.
`table`	a logical value indicating whether a summary table with the regression results should be printed.
`variance`	a logical value indicating whether the variance of the estimated parameters should be calculated. Defaults to `TRUE` whenever `table = TRUE`.
`var.type`	a description of the type of variance of the estimated parameters to be calculated. Options are `standard` (recommended for models estimated by maximum likelihood, such as the binary component of two-part models), `robust` (recommended for models estimated by quasi-maximum likelihood, such as standard fractional regression models or the fractional component of a two-part fractional regression model), `cluster` (recommended in the case of panel data) and `default` (implements the `standard` or `robust` versions as appropriate).

`var.eim`	a logical value indicating whether the expected information matrix should be used in the calculation of the variance. When false, the observation information matrix will be used. Defaults to `TRUE`.
`var.cluster`	a numeric vector containing the values of the variable that specifies to which cluster each observation belongs.
`dfc`	a logical value indicating whether a degrees of freedom correction should be applied to the covariance matrix. Defaults to `FALSE`.
`...`	Arguments to pass to glm.

Details

`fracreg` estimates one-part, two-part hurdle, and three-part double-inflated fractional regression models; see Ramalho, Ramalho and Murteira (2011) and Fang and Ma (2013) for details on those models. The one-part models and the fractional component of two- and three-part models are estimated by Bernoulli-based quasi-maximum likelihood, while the binary components of two- and three-part models are estimated by maximum likelihood. `fracreg` uses the standard glm command to perform the estimations. Therefore, `fracreg` is essentially a convenience command, allowing estimation of several alternative fractional regression models using the same command. In addition, `fracreg` provides an R-squared measure for all models (calculated as the square of the correlation coefficient between the actual and fitted values of the dependent variable), calculates the fitted values of the dependent variable in two-part models and stores the information needed to implement some very useful commands for fractional regression models: fracreg.reset (RESET test), fracreg.ptest (P test), fracreg.ggoff (GGOFF tests) and fracreg.pe (partial effects).

Value

When `type = "1P"` or `"2Pfrac"`, `fracreg` returns a list with the following elements:

`class`	"fracreg".
`formula`	the model formula.
`type`	the name of the estimated model.
`link`	the name of the specified link.
`method`	estimation method. Currently, "QML" (quasi-maximum likelihood) for fractional components or models and"ML" (maximum likelihood) for the binary component of two-part models.
`p`	a named vector of coefficients.
`yhat`	the fitted mean values.
`xbhat`	the fitted mean values of the linear predictor.
`converged`	logical. Was the algorithm judged to have converged?
`x.names`	a vector containing the names of the covariates.

If `variance = TRUE` or `table = TRUE`, the previous list also contains the following elements:

`p.var`	a named covariance matrix.
`var.type`	covariance matrix type.
`var.eim`	logical. Was the expected information matrix used in the computation of the covariance matrix?
`dfc`	logical. Was a degrees of freedom correction used for the computation of the covariance matrix?

If `var.type = "cluster"`, the list also contains the following element:

`var.cluster`	the variable that specifies to which cluster each observation belongs.

When `type = "2Pbin"`, `fracreg` returns a similar list with the following additional element:

`LL`	the value of the log-likelihood.

When `type = "2P"`, `fracreg` returns the previous lists, indexed by the prefixes `resBIN` and `resFRAC`, and the following additional elements:

`class`	"fracreg".
`type`	"2P".
`ybase`	a numeric vector containing the values of the response variable.
`x2base`	a numeric matrix containing the values of the covariates.
`yhat2P`	the overall fitted mean values.
`converged`	logical. Were the algorithms judged to have converged in both parts of the model?

When `type = "3P"`, `fracreg` returns the previous lists, indexed by the prefixes `resBIN0`, `resBIN1`, and `resFRAC`, and the following additional elements:

`class`	"fracreg".
`type`	"3P".
`ybase`	a numeric vector containing the values of the response variable.
`x2base`	a numeric matrix containing the values of the covariates.
`yhat3P`	the overall fitted mean values.
`converged`	logical. Were the algorithms judged to have converged in all parts of the model?

Author(s)

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References

Papke, L. E. and Wooldridge, J. M. (1996), "Econometric methods for fractional response variables with an application to 401(k) plan participation rates", *Journal of Applied Econometrics*, 11(6), 619-632.

Ramalho, E.A., J.J.S. Ramalho and J.M.R. Murteira (2011), "Alternative estimating and testing empirical strategies for fractional regression models", *Journal of Economic Surveys*, 25(1), 19-68.

Fang, K., & Ma, S. (2013), "Three-part model for fractional response variables with application to Chinese household health insurance coverage", *Journal of Applied Statistics*, 40(5), 925-940.

See Also

`fracreg.reset` and `fracreg.ggoff`, for specification tests.
`fracreg.ptest`, for non-nested hypothesis tests.
`fracreg.pe`, for computing partial effects.
`fracreghet`, for fitting cross-sectional fractional regression models with unobserved heterogeneity.
`fracregpd`, for fitting panel data fractional regression models.

Examples

```
N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

ym <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))
y <- rbeta(N,ym*20,20*(1-ym))
y[y > 0.9] <- 1

#fracreg estimation of a logit fractional regression model
fracreg(y,X,linkfrac="logit")

#fracreg estimation of the binary logit component of the two-part fractional
#regression model with y=1 as the relevant boundary value
fracreg(y,X,linkbin="logit",type="2Pbin",inf=1)

#fracreg estimation of the fractional component of the two-part fractional
#regression model with y=1 as the relevant boundary value and using a
#probit link function
fracreg(y,X,linkfrac="probit",type="2Pfrac",inf=1)

#fracreg estimation of both components of a two-part fractional regression model
#with y=1 as the relevant boundary value and using a cloglog binary link
#function and a logit fractional link function
fracreg(y,X,linkbin="cloglog",linkfrac="logit",type="2P",inf=1)

#Three-part model (y has both 0s and 1s)
y3p <- y
y3p[1:20] <- 0
y3p[21:40] <- 1
fracreg(y3p,X,linkbin=c("logit","probit"),linkfrac="logit",type="3P")
```

`fracreg.ggoff` *GGOFF Tests for Fractional Regression Models*

Description

`fracreg.ggoff` is used to perform Generalized Goodness-Of-Functional-Form (GGOFF) tests to check the adequacy of the functional form and link specification of fractional regression models.

Usage

```
fracreg.ggoff(object, version = "LM", table = TRUE, ...)
```

Arguments

`object`	an object containing the results of an `fracreg` command.
`version`	a vector containing the test versions to use. Available options: `Wald`, `LM` (the default) and, only for the binary component of two-part models, `LR`. More than one option may be chosen.

`table`	a logical value indicating whether a summary table with the test results should be printed.
`...`	Arguments to pass to glm, which is used to estimate the model under the alternative hypothesis when `version` is a vector containing `"Wald"` or `"LR"`.

Details

`fracreg.ggoff` applies the GGOFF, GOFF1 and GOOFF2 test statistics to fractional regression models estimated via `fracreg`. `fracreg.ggoff` may be used to test the link specification of: (i) one-part fractional regression models; (ii) the binary component of two-part fractional regression models; and (iii) the fractional component of two-part fractional regression models. When the `Wald` version is implemented, it is taken into account the option that was chosen for computing standard errors in the model under evaluation. For the `LM` version, a robust version is computed in cases (i) and (iii) and a conventional version in case (ii). See Ramalho, Ramalho and Murteira (2014) for details on the application of the GGOFF, GOFF1 and GOOFF2 tests in the fractional regression framework.

Value

`fracreg.ggoff` returns a named vector with the test results.

Author(s)

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References

Ramalho, E.A., J.J.S. Ramalho and J.M.R. Murteira (2014), "A generalized goodness-of-functional form test for binary and fractional regression models", *Manchester School*, 82(4), 488-507.

See Also

`fracreg`, for fitting fractional regression models.
`fracreg.reset`, for asymptotically equivalent specification tests.
`fracreg.ptest`, for non-nested hypothesis tests.
`fracreg.pe`, for computing partial effects.

Examples

```
N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

ym <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))
y <- rbeta(N,ym*20,20*(1-ym))
y[y > 0.9] <- 1

#Testing the logit specification of a standard fractional regression model
#using LM and Wald versions of the GGOFF test, based on 1 or 2 fitted powers of
#the linear predictor
res <- fracreg(y,X,linkfrac="logit",table=FALSE)
fracreg.ggoff(res,c("Wald","LM"))
```

```
#Testing the probit specification of the binary component of a two-part fractional
#regression model using a LR-based GGOFF test
res <- fracreg(y,X,linkbin="probit",type="2Pbin",inf=1,table=FALSE)
fracreg.ggoff(res,"LR")
```

`fracreg.pe` *Fractional Regression Models - Partial Effects*

Description

`fracreg.pe` is used to compute average and/or conditional partial effects in fractional regression models.

Usage

```
fracreg.pe(object, APE = TRUE, CPE = FALSE, at = NULL, which.x = NULL,
           variance = TRUE, table = TRUE)
```

Arguments

`object`	an object containing the results of an `fracreg` command.
`APE`	a logical value indicating whether average partial effects are to be computed.
`CPE`	a logical value indicating whether conditional partial effects are to be computed.
`at`	a numeric vector containing the covariates' values at which the conditional partial effects are to be computed or the strings `"mean"` (the default) or `"median"`, in which cases the covariates are evaluated at their mean or median values (or mode, in case of dummy variables), respectively.
`which.x`	a vector containing the names of the covariates to which the partial effects are to be computed.
`variance`	a logical value indicating whether the variance of the estimated partial effects should be calculated. Defaults to `TRUE` whenever `table` = `TRUE`.
`table`	a logical value indicating whether a summary table with the results should be printed.

Details

`fracreg.pe` calculates partial effects for fractional regression models estimated via `fracreg`. `fracreg.pe` may be used to compute average or conditional partial effects for: (i) one-part fractional regression models; (ii) the binary components of two-part and three-part fractional regression models; (iii) the fractional components of two-part and three-part fractional regression models; and (iv) two-part and three-part fractional regression models overall. For calculating standard errors, it is taken into account the option that was previously chosen for estimating the model. See Ramalho, Ramalho and Murteira (2011) and Fang and Ma (2013) for details on the computation of partial effects in the fractional regression framework.

Value

`fracreg.pe` returns a list with the following element:

PE.p — a named vector of partial effects.

If `variance = TRUE` or `table = TRUE`, the previous list also contains the following element:

PE.sd — a named vector of standard errors of the estimated partial effects.

When both average and conditional partial effects are requested, two lists containing the previous elements are returned, indexed by the prefixes ape and cpe.

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References

Ramalho, E.A., J.J.S. Ramalho and J.M.R. Murteira (2011), "Alternative estimating and testing empirical strategies for fractional regression models", *Journal of Economic Surveys*, 25(1), 19-68.

Fang, K., & Ma, S. (2013), "Three-part model for fractional response variables with application to Chinese household health insurance coverage", *Journal of Applied Statistics*, 40(5), 925-940.

See Also

`fracreg`, for fitting fractional regression models.
`fracreg.reset` and `fracreg.ggoff`, for specification tests.
`fracreg.ptest`, for non-nested hypothesis tests.

Examples

```
N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

ym <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))
y <- rbeta(N,ym*20,20*(1-ym))
y[y > 0.9] <- 1

#Computing average partial effects for a logit fractional regression model
res <- fracreg(y,X,linkfrac="logit",table=FALSE)
fracreg.pe(res)

#Computing average partial effects for a binary logit + fractional probit
#two-part model
res <- fracreg(y,X,linkbin="logit",linkfrac="probit",type="2P",inf=1,table=FALSE)
fracreg.pe(res)

#Computing conditional partial effects for X2 in the logit component
#of a two-part fractional regression model, with the covariates evaluated
#at their median values
res <- fracreg(y,X,linkfrac="logit",type="2Pfrac",inf=1,table=FALSE)
fracreg.pe(res,APE=FALSE,CPE=TRUE,at="median",which.x="X2")
```

```
#Computing average partial effects for a three-part double-inflated model
y3p <- y
y3p[1:20] <- 0
y3p[21:40] <- 1
res3p <- fracreg(y3p,X,linkbin=c("logit","probit"),linkfrac="logit",type="3P",table=FALSE)
fracreg.pe(res3p)
```

---

fracreg.ptest — *P Test for Fractional Regression Models*

Description

`fracreg.ptest` is used to perform the P test to evaluate the specification of alternative, non-nested fractional regression models by testing against each other.

Usage

```
fracreg.ptest(object1, object2, version = "Wald", table = TRUE)
```

Arguments

`object1`	an object containing the results of an `fracreg` command.
`object2`	an object containing the results of another `fracreg` command.
`version`	a vector containing the test versions to use. Available options: `Wald` (the default) and `LM`. Both options may be chosen at the same time and are computed in a robust way.
`table`	a logical value indicating whether a summary table with the test results should be printed.

Details

`fracreg.ptest` applies the P test statistic proposed by Davidson and MacKinnon (1981) to fractional regression models estimated via `fracreg`. `fracreg.ptest` may be used to test against each other two alternative specifications for the link function in: (i) one-part fractional regression models; (ii) the binary components of two-part and three-part fractional regression models; (iii) the fractional components of two-part and three-part fractional regression models; and (iv) two-part and three-part fractional regression models. In addition, `fracreg.ptest` may be used to test one-part models against two-part or three-part models and in cases where the link functions are the same but the regressors are non-nested. See Ramalho, Ramalho and Murteira (2011) for details on the application of the P test in the fractional regression framework.

Value

`fracreg.reset` returns a named vector with the test results.

Author(s)

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References

Davidson, R. and J.G. MacKinnon (1981), "Several tests for model specification on the presence of alternative hypotheses", *Econometrica*, 49(3), 781-793.

Ramalho, E.A., J.J.S. Ramalho and J.M.R. Murteira (2011), "Alternative estimating and testing empirical strategies for fractional regression models", *Journal of Economic Surveys*, 25(1), 19-68.

See Also

`fracreg`, for fitting fractional regression models.
`fracreg.reset` and `fracreg.ggoff`, for specification tests.
`fracreg.pe`, for computing partial effects.

Examples

```
N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

ym <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))
y <- rbeta(N,ym*20,20*(1-ym))
y[y > 0.9] <- 1

#Testing logit versus loglog specifications for standard fractional
#regression models using a LM version of the P test
res1 <- fracreg(y,X,linkfrac="logit",table=FALSE)
res2 <- fracreg(y,X,linkfrac="loglog",table=FALSE)
fracreg.ptest(res1,res2,"LM")

#Testing a logit one-part fractional regression model versus a binary logit +
#fractional probit two-part model using a Wald version of the P test
res1 <- fracreg(y,X,linkfrac="logit",table=FALSE)
res2 <- fracreg(y,X,linkbin="logit",linkfrac="probit",type="2P",inf=1,table=FALSE)
fracreg.ptest(res1,res2,"Wald")
```

`fracreg.reset` *RESET Test for Fractional Regression Models*

Description

`fracreg.reset` is used to perform the Regression Equation Specification Error Test (RESET) to check the functional form and specification of fractional regression models.

Usage

```
fracreg.reset(object, lastpower.vec = 3, version = "LM", table = TRUE, ...)
```

Arguments

`object`	an object containing the results of an `fracreg` command.
`lastpower.vec`	a numeric vector containing the maximum powers of the linear predictors to be used in RESET tests.
`version`	a vector containing the test versions to use. Available options: `Wald`, `LM` (the default) and, only for the binary component of two-part models, `LR`. More than one option may be chosen.
`table`	a logical value indicating whether a summary table with the test results should be printed.
`...`	Arguments to pass to glm, which is used to estimate the model under the alternative hypothesis when `version` is a vector containing `"Wald"` or `"LR"`.

Details

`fracreg.reset` applies the RESET test statistic to fractional regression models estimated via `fracreg`. `fracreg.reset` may be used to test the link specification of: (i) one-part fractional regression models; (ii) the binary components of two-part and three-part fractional regression models; and (iii) the fractional components of two-part and three-part fractional regression models. When the `Wald` version is implemented, it is taken into account the option that was chosen for computing standard errors in the model under evaluation. For the `LM` version, a robust version is computed in cases (i) and (iii) and a conventional version in case (ii). See Ramalho, Ramalho and Murteira (2011) for details on the application of the RESET test in the fractional regression framework.

Value

`fracreg.reset` returns a named vector with the test results.

Author(s)

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References

Ramalho, E.A., J.J.S. Ramalho and J.M.R. Murteira (2011), "Alternative estimating and testing empirical strategies for fractional regression models", *Journal of Economic Surveys*, 25(1), 19-68.

See Also

`fracreg`, for fitting fractional regression models.
`fracreg.ggoff`, for asymptotically equivalent specification tests.
`fracreg.ptest`, for non-nested hypothesis tests.
`fracreg.pe`, for computing partial effects.

Examples

```
N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

ym <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))
y <- rbeta(N,ym*20,20*(1-ym))
```

```
y[y > 0.9] <- 1

#Testing the logit specification of a standard fractional regression model
#using LM and Wald versions of the RESET test, based on 1 or 2 fitted powers of
#the linear predictor
res <- fracreg(y,X,linkfrac="logit",table=FALSE)
fracreg.reset(res,2:3,c("Wald","LM"))

#Testing the probit specification of the binary component of a two-part fractional
#regression model using LR-based RESET tests with quadratic and cubic fitted
#powers of the linear predictor
res <- fracreg(y,X,linkbin="probit",type="2Pbin",inf=1,table=FALSE)
fracreg.reset(res,3,"LR")
```

---

fracreghet *Fitting Fractional Regression Models under Unobserved Heterogeneity*

Description

fracreghet is used to fit fractional regression models under unobserved heterogeneity, i.e. regression models for proportions, percentages or fractions that suffer from neglected heterogeneity and/or endogeneity issues.

Usage

```
fracreghet(y, x, z = x, var.endog, start, type = "GMMx", link = "logit",
           intercept = TRUE, table = TRUE, variance = TRUE,
           var.type = "robust", var.cluster, adjust = 0, ...)
```

Arguments

y	a numeric vector containing the values of the response variable.
x	a numeric matrix, with column names, containing the values of all covariates (exogenous and endogenous).
z	a numeric matrix, with column names, containing the values of all exogenous variables (covariates and instrumental variables). Defaults to x.
var.endog	a numeric vector containing the values of the endogenous covariate (or of some transformation of it), which will be used as dependent variable in the linear reduced form assumed for application of xv-type estimators.
start	a numeric vector containing the initial values for the parameters to be optimized. Optional.
type	a description of the estimator to compute: GMMx (the default), GMMxv, GMMz, LINx, LINxv, LINz or QMLxv.
link	a description of the link function to use. Available options for all estimators: logit and cloglog. Additional available options for QML and LIN estimators: probit, cauchit and loglog.
intercept	a logical value indicating whether the model should include a constant term or not.

`table`	a logical value indicating whether a summary table with the regression results should be printed.
`variance`	a logical value indicating whether the variance of the estimated parameters should be calculated. Defaults to `TRUE` whenever `table = TRUE`.
`var.type`	a description of the type of variance of the estimated parameters to be calculated. Options are `robust`, the default, and `cluster`.
`var.cluster`	a numeric vector containing the values of the variable that specifies to which cluster each observation belongs.
`adjust`	the numeric value to be added to the response variable in case of boundary observations when the LIN estimators are applied or the string `drop`, which implies that the boundary observations are dropped.
`...`	Arguments to pass to nlminb.

Details

`fracreghet` computes the GMM estimators proposed in Ramalho and Ramalho (2017) for fractional regression models with unobserved heterogeneity: GMMx, which allows for neglected heterogeneity but not for endogeneity; GMMxv, which allows both issues and assumes a linear reduced form for the endogeneous covariate (or for a transformation of it); and GMMz, which also allows for both issues but does not require the assumption of a reduced form for the endogenous covariate. In addition, `fracreghet` also computes three linearized estimators (LINx, LINxv and LINz) that have similar features to their GMM counterparts. It also provides a QML estimator (QMLxv) that addresses endogeneity using a Control Function (CF) approach, which includes the first-stage reduced-form residuals as an additional regressor in the main fractional equation, providing a Hausman-type test for endogeneity. See Ramalho and Ramalho (2017) for details on each estimator. For overidentified models, `fracreghet` calculates Hansen's J statistic. For `GMMx` and `LINx`, `fracreghet` stores the information needed to implement the RESET test (fracreghet.reset). For all estimators, `fracreghet` stores the information needed to calculate partial effects (fracreghet.pe).

Value

`fracreghet` returns a list with the following elements:

`class`	"fracreghet".
`formula`	the model formula.
`type`	the name of the estimator computed.
`link`	the name of the specified link.
`adjust`	The value or the type of the adjustment applied to LIN estimators.
`p`	a named vector of coefficients.
`Hy`	the transformed values of the response variable when GMM or LIN estimators are computed or the values of the response variable in the QML case.
`xbhat`	the fitted mean values of the linear predictor (for xv-type estimators, includes the term relative to the first-stage residual).
`converged`	logical. Was the algorithm judged to have converged?
`x.names`	a vector containing the names of the covariates.

In case of an overidentifying model, the following element is also returned:

`J`	the result of Hansen's J test of overidentifying moment conditions.

If `variance = TRUE` or `table = TRUE` and the algorithm converged successfully, the previous list also contains the following elements:

`p.var`	a named covariance matrix.
`var.type`	covariance matrix type.

If `var.type = "cluster"`, the list also contains the following element:

`var.cluster`	the variable that specifies to which cluster each observation belongs.

Author(s)

Sulman Olieko Owili <oliekosulman@gmail.com>

References

Ramalho, E. A., & Ramalho, J. J. S. (2017), "Moment-based estimation of nonlinear regression models with boundary outcomes and endogeneity, with applications to nonnegative and fractional responses", *Econometric Reviews*, 36(4), 397-420.

See Also

`fracreghet.reset`, for the RESET test.
`fracreghet.pe`, for computing partial effects.
`fracreg`, for fitting standard cross-sectional fractional regression models.
`fracregpd`, for fitting panel data fractional regression models.

Examples

```
N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

Z <- cbind(rnorm(N),rnorm(N),rnorm(N))
dimnames(Z)[[2]] <- c("Z1","Z2","Z3")

y <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))

#Exogeneity, GMMx estimator
fracreghet(y,X,type="GMMx")

#Endogeneity, GMMz estimator
fracreghet(y,X,Z,type="GMMz")

#Endogeneity, GMMxv estimator
fracreghet(y,X,Z,X[,1],type="GMMxv")
```

fracreghet.pe *Fractional Regression Models under Unobserved Heterogeneity - Partial Effects*

Description

fracreghet.pe is used to compute average and/or conditional partial effects in fractional regression models under unobserved heterogeneity.

Usage

```
fracreghet.pe(object, smearing = TRUE, APE = TRUE, CPE = FALSE, at = NULL,
              which.x = NULL, table = TRUE, variance = TRUE)
```

Arguments

object	an object containing the results of an fracreghet command.
smearing	a logical value indicating whether the smearing correction is to be applied
APE	a logical value indicating whether average partial effects are to be computed.
CPE	a logical value indicating whether conditional partial effects are to be computed.
at	a numeric vector containing the covariates' values at which the conditional partial effects are to be computed or the strings "mean" (the default) or "median", in which cases the covariates are evaluated at their mean or median values (or mode, in case of dummy variables), respectively.
which.x	a vector containing the names of the covariates to which the partial effects are to be computed.
table	a logical value indicating whether a summary table with the results should be printed.
variance	a logical value indicating whether the variance of the estimated partial effects should be calculated. Defaults to TRUE whenever table = TRUE.

Details

fracreghet.pe calculates partial effects for fractional regression models estimated via fracreghet. fracreghet.pe may be used to compute average or conditional partial effects. These partial effects may be conditional only on observables, using the smearing estimator, or also on unobservables, setting the error term to zero. For calculating standard errors, it is taken into account the option that was previously chosen for estimating the model. See Ramalho and Ramalho (2017) for details on the computation of partial effects for fractional regression models under unobserved heterogeneity.

Value

fracreghet.pe returns a list with the following element:

PE.p	a named vector of partial effects.

If variance = TRUE or table = TRUE, the previous list also contains the following element:

PE.sd	a named vector of standard errors of the estimated partial effects.

When both average and conditional partial effects are requested, two lists containing the previous elements are returned, indexed by the prefixes ape and cpe.

Author(s)

Sulman Olieko Owili <oliekosulman@gmail.com>

References

Ramalho, E. A., & Ramalho, J. J. S. (2017), "Moment-based estimation of nonlinear regression models with boundary outcomes and endogeneity, with applications to nonnegative and fractional responses", *Econometric Reviews*, 36(4), 397-420.

See Also

`fracreghet`, for fitting fractional regression models under unobserved heterogeneity.
`fracreghet.reset`, for the RESET test.

Examples

```
N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

Z <- cbind(rnorm(N),rnorm(N),rnorm(N))
dimnames(Z)[[2]] <- c("Z1","Z2","Z3")

y <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))

res <- fracreghet(y,X,type="GMMx",table=FALSE)

#Smearing estimator of average partial effects for variable X1
fracreghet.pe(res,which.x="X1")

#Naive estimator of conditional partial effects for all covariates,
#which are evaluated at X1=1 and X2=-1
fracreghet.pe(res,smearing=FALSE,APE=FALSE,CPE=TRUE,at=c(1,-1))
```

`fracreghet.reset` *RESET Test for Fractional Regression Models under Neglected Heterogeneity*

Description

`fracreghet.reset` is used to test the specification of fractional regression models estimated by GMMx or LINx.

Usage

```
fracreghet.reset(object, lastpower.vec = 3, version = "Wald", table = T, ...)
```

Arguments

object	an object containing the results of an fracreghet command.
lastpower.vec	a numeric vector containing the maximum powers of the linear predictors to be used in RESET tests.
version	a vector containing the test versions to use. Available options: Wald (the default) and LM (only available for GMMx).
table	a logical value indicating whether a summary table with the test results should be printed.
...	Arguments to pass to nlminb, which is used to estimate the model under the alternative hypothesis when version is equal to "Wald" and the null model was estimated by GMMx.

Details

fracreghet.reset applies the RESET test statistic to fractional regression models estimated via fracreghet using the options GMMx or LINx. fracreghet.reset may be used to test simultaneously the validity of the link specification and the transformation applied to the response variable by each estimator. It is taken into account the option that was chosen for computing standard errors in the model under evaluation. See Ramalho and Ramalho (2017) for details.

Value

fracreghet.reset returns a named vector with the test results.

Author(s)

Sulman Olieko Owili <oliekosulman@gmail.com>

References

Ramalho, E. A., & Ramalho, J. J. S. (2017), "Moment-based estimation of nonlinear regression models with boundary outcomes and endogeneity, with applications to nonnegative and fractional responses", *Econometric Reviews*, 36(4), 397-420.

See Also

fracreghet, for fitting fractional regression models under unobserved heterogeneity.
fracreghet.pe, for computing partial effects.

Examples

```
N <- 250
u <- rnorm(N)

X <- cbind(rnorm(N),rnorm(N))
dimnames(X)[[2]] <- c("X1","X2")

Z <- cbind(rnorm(N),rnorm(N),rnorm(N))
dimnames(Z)[[2]] <- c("Z1","Z2","Z3")

y <- exp(X[,1]+X[,2]+u)/(1+exp(X[,1]+X[,2]+u))

res <- fracreghet(y,X,type="GMMx",table=FALSE)
```

```
#LM and Wald versions of the RESET test, based on 1 or 2 fitted powers of xb
fracreghet.reset(res,2:3,c("Wald","LM"))
```

---

fracregpd *Fitting Panel Data Fractional Regression Models*

Description

`fracregpd` is used to fit panel data regression models when the dependent variable has a bounded, fractional nature.

Usage

```
fracregpd(id, time, y, x, z, var.endog, x.exogenous = T, lags, start, type,
 GMMww.cor = T, link = "logit", intercept = T, table = T, variance = T,
 var.type = "cluster", tdummies = F, bootstrap = F, B = 200, ...)
```

Arguments

`id`	a numeric vector identifying the cross-sectional units.
`time`	a numeric vector identifying the time periods in which the cross-sectional units were observed.
`y`	a numeric vector containing the values of the response variable.
`x`	a numeric matrix, with column names, containing the values of all covariates (exogenous and endogenous).
`z`	a numeric matrix, with column names, containing the values of all exogenous variables (covariates and external instrumental variables). Only required in case of endogenous explanatory variables.
`var.endog`	a numeric vector containing the values of the endogenous covariate (or of some transformation of it), which will be used as dependent variable in the linear reduced form assumed for application of the `QMLcre` estimator. Only required for this estimator.
`x.exogenous`	a logical value indicating whether all explanatory variables are assumed to be exogenous or not.
`lags`	a logical value indicating whether the first lags of `x` or `z` should be used as instruments for `x`. Defaults to `TRUE` for the GMMww and GMMc estimators and to `FALSE` for the remaining estimators. The `GMMcre` and `QMLcre` estimators do not admit lagged instruments.
`start`	a numeric vector containing the initial values for the parameters to be optimized. Optional.
`type`	a description of the estimator to compute: `GMMww`, `GMMc`, `GMMbgw`, `GMMpfe`, `GMMcre`, `GMMpre` or `QMLcre`.
`GMMww.cor`	a logical value indicating whether each explanatory variable should be transformed in deviations from its overall mean before computing the `GMMww` estimator.
`link`	a description of the link function to use. Available options for all GMM estimators: `logit` and `cloglog`. Only option for the `QMLcre` estimator: `probit`.

intercept	a logical value indicating whether the model should include a constant term or not. Only relevant for the GMMpre estimator.
table	a logical value indicating whether a summary table with the regression results should be printed.
variance	a logical value indicating whether the variance of the estimated parameters should be calculated. Defaults to TRUE whenever table = TRUE.
var.type	a description of the type of variance of the estimated parameters to be calculated. Options are cluster, the default, and robust. In overidentified models, it also affects the parameter estimates via the GMM weighting matrix.
tdummies	a logical value indicating whether time dummies should be included among the model explanatory variables.
bootstrap	a logical value indicating whether bootstrap should be used in the estimation of the parameter standard errors.
B	the number of bootstrap replications.
...	Arguments to pass to nlminb.

Details

fracregpd computes the GMM estimators proposed in Ramalho, Ramalho and Coelho (2018) for panel data fractional regression models with both time-variant and time-invariant unobserved heterogeneity and endogeneous covariates: GMMww, GMMc, GMMbgw, GMMpfe, GMMcre and GMMpre. In addition, fracregpd also computes QMLcre, which was proposed by Papke and Wooldridge (2008) and Wooldridge (2010). The Correlated Random Effects (CRE) approach models the unobserved individual-specific effects as a linear function of the time averages of covariates, allowing consistent estimation even when unobserved effects are correlated with regressors. For overidentified models, fracregpd calculates Hansen's J statistic.

Value

fracregpd returns a list with the following elements:

type	the name of the estimator computed.
link	the name of the specified link.
p	a named vector of coefficients.
Hy	the transformed values of the response variable when GMM estimators are computed or the values of the response variable in the QML case.
converged	logical. Was the algorithm judged to have converged?

In case of an overidentifying model, the following element is also returned:

J	the result of Hansen's J test of overidentifying moment conditions.

If variance = TRUE or table = TRUE and the algorithm converged successfully, the previous list also contains the following elements:

p.var	a named covariance matrix.
var.type	covariance matrix type.

Author(s)

Sulman Olieko Owili <oliekosulman@gmail.com>

References

Papke, L. and Wooldridge, J.M. (2008), "Panel data methods for fractional response variables with an application to test pass rates", *Journal of Econometrics*, 145(1-2), 121-233.

Ramalho, E. A., Ramalho, J. J. S., & Coelho, L. M. S. (2018), "Exponential Regression of Fractional-Response Fixed-Effects Models with an Application to Firm Capital Structure", *Journal of Econometric Methods*, 7(1), 20150019.

Wooldridge, J.M. (2010), "Correlated random effects models with unbalanced panels", mimeo.

See Also

fracreg, for fitting standard cross-sectional fractional regression models.
fracreghet, for fitting cross-sectional fractional regression models with unobserved heterogeneity.

Examples

```
id <- rep(1:50,each=5)
time <- rep(1:5,50)
NT <- 250

XBu <- rnorm(NT)
y <- exp(XBu)/(1+exp(XBu))

X <- cbind(rnorm(NT),rnorm(NT))
dimnames(X)[[2]] <- c("X1","X2")

Z <- cbind(rnorm(NT),rnorm(NT),rnorm(NT))
dimnames(Z)[[2]] <- c("Z1","Z2","Z3")

# Exogeneity, no lags, no time dummies, clustered standard errors, GMMbgw estimator
fracregpd(id,time,y,X,type="GMMbgw")

# Lagged covariates and instruments, robust standard errors, GMMww estimator
fracregpd(id,time,y,X,lags=TRUE,type="GMMww",var.type="robust")

# Endogeneity, time dummies, GMMpfe estimator
fracregpd(id,time,y,X,Z,x.exogenous=FALSE,type="GMMpfe",tdummies=TRUE)

# Standard errors based on 100 bootstrap samples, QMLcre estimator (not run)
#fracregpd(id,time,y,X,Z,X[,1],x.exogenous=FALSE,type="QMLcre",link="probit",bootstrap=TRUE,B=100)
```

Index